- **Objective and scope:**
    - Analyze Splunk tutorial SSH logs to investigate repeated failed logins from 194.8.74.23, identify targeted hosts and usernames, and search for successful logins from the same IP. Summarize the evidence in a dashboard and investigation report. This lab uses practice data, not a live incident.
- **Tools and dataset:**
    - *Tool*: Splunk Cloud Platform trial, Search & Reporting.
    - *Dataset*: Splunk's official **tutorialdata.zip**.
    - *Log type investigated:* **secure-2** SSH authentication logs.
    - *Search time range*: All time.
    - *Dataset counts:* 39,532 web access events, 40,088 security events, and 30,244 sales events.
    - *Scope:* Investigation focused on source IP **194.8.74.23.**
- **Searches performed:**
    - 1) Count failed logins by host for the investigated IP
        - source=*tutorialdata.zip:* sourcetype="secure-2" "194.8.74.23" "Failed password"
          | stats count by host
          | sort - count
        - **Purpose**: Identify which hosts recorded failed login attempts matching this IP.
    - 2) Search for successful SSH logins from the investigated IP
        - source=*tutorialdata.zip:* sourcetype="secure-2" "194.8.74.23" ("Accepted password" OR "Accepted publickey")
        - **Purpose**: Check for successful password or public-key authentication matching this IP.
        - **Result**: No matching events were found with All time selected.
    - 3) Identify targeted usernames
        - source=*tutorialdata.zip:* sourcetype="secure-2" "194.8.74.23" "Failed password"
          | rex "Failed password for (?:invalid user)?(?<target_user>\S+) from"
          | stats count by target_user
          | sort - count
        - **Purpose**: Extract attempted usernames and count their failures. The dashboard adds | head 10 to display only the top 10.
- **Findings**:
    - The security logs contained 33,253 failed-password events across all source IPs.
    - 132 failures matched **194.8.74.23**: www1 (62), mailsv (45), www3 (19), and www2 (6).
    - The most frequently targeted usernames were local (6), system (6), and irc (5).

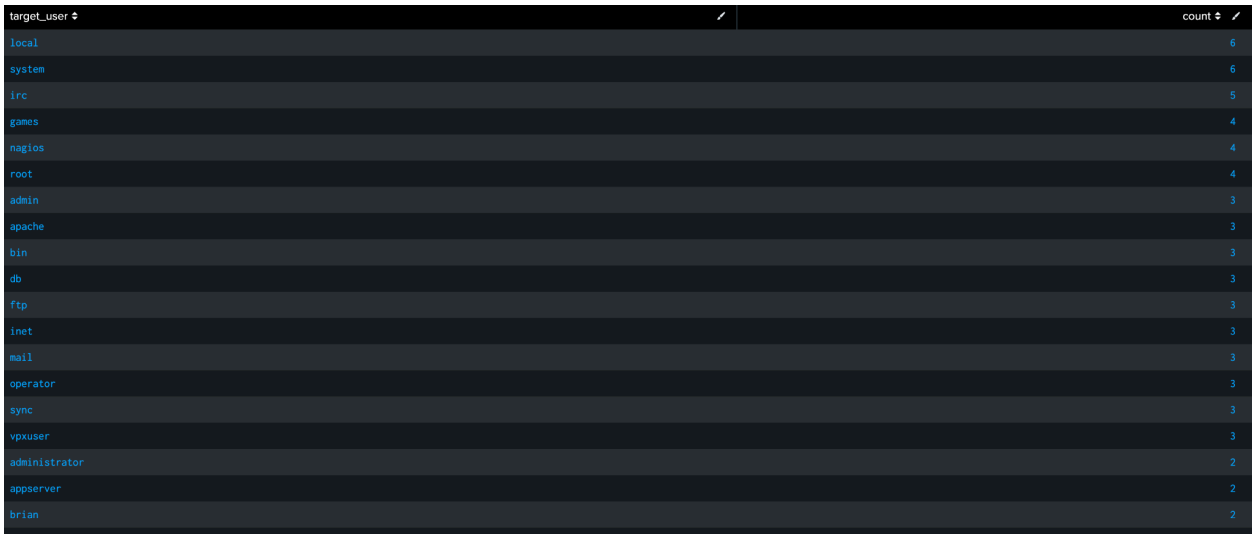
- No successful password or public-key SSH logins matching this IP were found.
- Multiple targeted accounts and hosts suggest automated login guessing. Successful compromise was not established.

- **Dashboard and screenshots:**
  - Sample failed SSH logins showing attempted usernames and recorded timestamps.



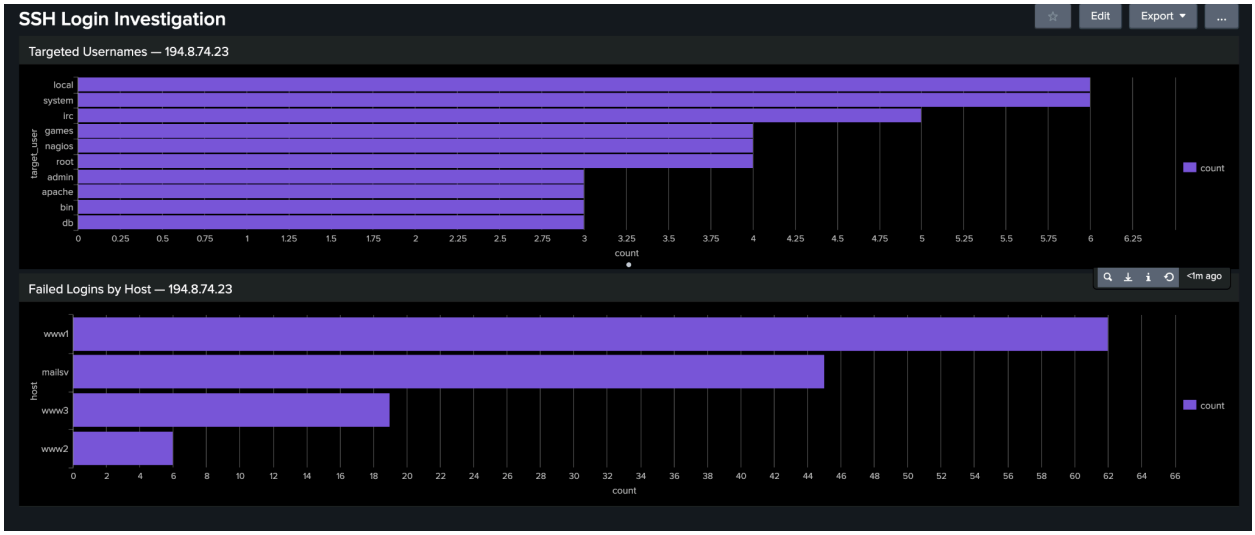
_time	host	_raw
2026-10-07 13:50:49	www2	Thu Oct 07 2026 13:50:49 www2 sshd[2989]: Failed password for invalid user workshop from 194.8.74.23 port 4108 ssh2
2026-10-07 13:50:49	www2	Thu Oct 07 2026 13:50:49 www2 sshd[1730]: Failed password for invalid user inet from 194.8.74.23 port 1609 ssh2
2026-10-07 13:50:49	www2	Thu Oct 07 2026 13:50:49 www2 sshd[4561]: Failed password for invalid user uni from 194.8.74.23 port 3976 ssh2
2026-10-07 13:50:49	www2	Thu Oct 07 2026 13:50:49 www2 sshd[3388]: Failed password for apache from 194.8.74.23 port 4370 ssh2
2026-10-07 13:50:49	www2	Thu Oct 07 2026 13:50:49 www2 sshd[3548]: Failed password for invalid user proxy from 194.8.74.23 port 3868 ssh2
2026-10-07 13:50:49	www2	Thu Oct 07 2026 13:50:49 www2 sshd[3990]: Failed password for mail from 194.8.74.23 port 2880 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[1719]: Failed password for ftp from 194.8.74.23 port 2608 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[3518]: Failed password for mail from 194.8.74.23 port 4546 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[2481]: Failed password for invalid user administrator from 194.8.74.23 port 4773 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[3732]: Failed password for invalid user system from 194.8.74.23 port 1471 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[4744]: Failed password for invalid user customer from 194.8.74.23 port 4475 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[1048]: Failed password for invalid user admin from 194.8.74.23 port 3513 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[5489]: Failed password for root from 194.8.74.23 port 4439 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[4526]: Failed password for news from 194.8.74.23 port 1334 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[3726]: Failed password for invalid user system from 194.8.74.23 port 2866 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[4321]: Failed password for nagios from 194.8.74.23 port 4559 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[5555]: Failed password for invalid user desktop from 194.8.74.23 port 1676 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[4034]: Failed password for invalid user perl from 194.8.74.23 port 2243 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[5331]: Failed password for bin from 194.8.74.23 port 1026 ssh2
2026-10-08 13:50:46	www1	Fri Oct 08 2026 13:50:46 www1 sshd[3249]: Failed password for invalid user postgres from 194.8.74.23 port 1890 ssh2

  - Below are the Failed-login counts grouped by attempted username for 194.8.74.23.

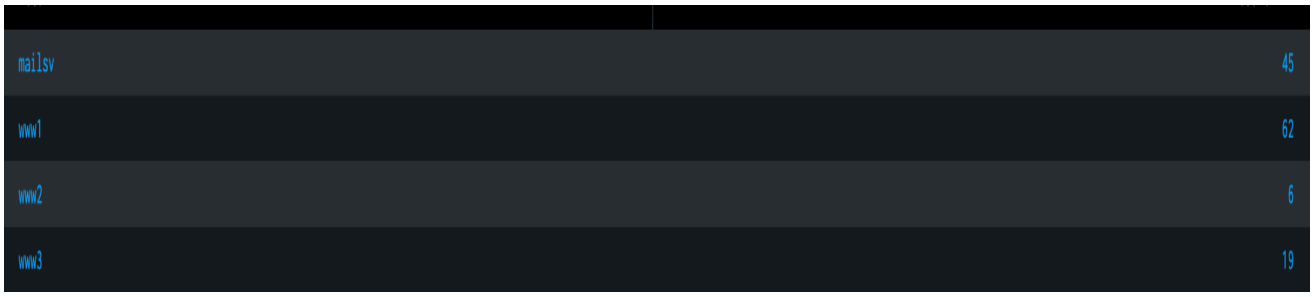


target_user	count
local	6
system	6
irc	5
games	4
nagios	4
root	4
admin	3
apache	3
bin	3
db	3
ftp	3
inet	3
mail	3
operator	3
sync	3
vpxuser	3
administrator	2
appserver	2
brian	2

- Below is the dashboard showing the top 10 targeted usernames and failed logins by host.



- Below is the image showing 132 failed-login events matching 194.8.74.23 across four hosts.



- **Verdict and limitations:**
  - The reviewed logs suggest that **194.8.74.23** repeatedly tried to guess login details on four hosts. No successful logins from that IP were found, so we cannot confirm a break-in. These findings apply only to the practice logs reviewed. Repeated timestamps make the attempt speed unclear, and the passwords tried are not shown.
- **Recommended response:**
  - In a real world incident/professional environment, it would be key to preserve the logs and check whether the activity was authorized. It would also be highly suggested to restrict SSH access to approved users and networks, consider blocking the source IP, and use login rate limits to slow repeated guesses. Reviewing successful logins and host activity for signs of a break-in would also be beneficial for the system's integrity. The statements provided are recommendations, as no response actions were performed in this lab.